

Computer Games Exercises: 2026s Phase B

<https://www.umm.uni-heidelberg.de/miism/>

Workflow

- The tasks are assigned to the groups in the exercise session on 06-17.
- One small project should be solved in small groups with 2 or 3 members. Every group should provide its own answer.
- The answers should be submitted to the corresponding Moodle assignment as one zip of all and only source files with proper names. (Moodle operation "Add submission")
- Zip file name: "CG_2026s_{LastName}_{GroupName}.zip"
- **All group members** should submit a short plan before the **due time** on 06-21.
- **All group members** should submit the final answer before the **due time** on 07-05.
- The answers are graded according to the performance. (Moodle operation "Grade")
- Every group should book one presentation slot on 07-08 or 07-15. (Moodle operation "Book slot")
- **All group members** should present in the presentation slot.

Tasks

Here are the tasks for selection.

- All games should use at least 3 techniques mentioned in the lecture.
- One recently developed technique listed below should be included.
- There should be one clearly organized report to explain the techniques with references.

Topic	Description
Game engine	a sophisticated agent model to let the characters operate with intelligence
Graphics 1	one recent tool to synthesize models and scenarios via AI
Graphics 2	a very fancy illumination via raytracing
Collision detection	AI based collision detection
Animation 1	realistic physics with rigid objects or soft objects
Animation 2	realistic physics with hydrodynamics
Path planning	a very sophisticated path planning